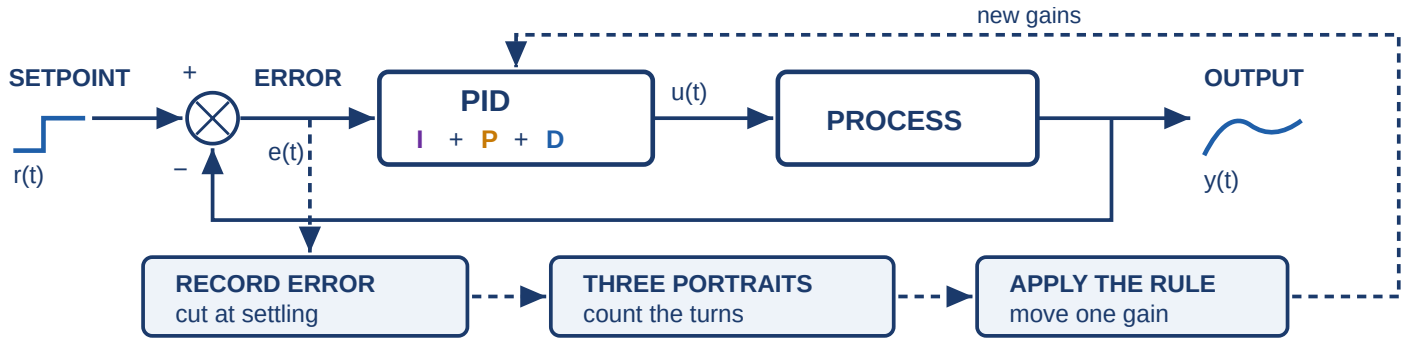


# SPIN TUNING

## STEP-RESPONSE PHASE-PORTRAIT INSPECTION

Read one closed-loop step response, move one gain, repeat. No model, no relay test, no optimizer.



$$u(t) = K_i \int e(\tau) d\tau + K_p e(t) + K_d \frac{de(t)}{dt}$$

each gain multiplies one signal — and each portrait plots that same signal; derivative filtered,  $T_f = K_d/\nu$ ,  $\nu = 10$

$K_i$  Integral  $\rightarrow \Gamma_0$

$K_p$  Proportional  $\rightarrow \Gamma_1$

$K_d$  Derivative  $\rightarrow \Gamma_2$

## WHAT EACH PORTRAIT SHOWS

$K_i$   $K_p$   $K_d$  = band (outer arc)    red = over limit, cut    green = within limit, raise

| PORTRAIT                                               | STEP RESPONSE                                                                     | PACHNER PLOTS — OVER / WITHIN | READING                                                                                                                                                                                                                             |
|--------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Gamma_0$<br>$K_i$ band<br>$\int e$ vs $e$            |                                                                                   | <br>$N_0 = 2.41$ $N_0 = 0.12$ | <ul style="list-style-type: none"><li>✓ Slow cycling — the loop hunts around setpoint</li><li>✓ Limit <math>\tilde{N}_0 = 0.50 = 2</math> quarter turns</li><li>✓ Over limit: cut <math>K_i</math> ↓, raise bands below ↑</li></ul> |
| $\Gamma_1$<br>$K_p$ band<br>$e$ vs $\Delta e$          | <br>$(K_i, K_p, K_d) = (0.35, 0.05, 0.5)$<br>$(K_i, K_p, K_d) = (0.2, 0.02, 0.7)$ | <br>$N_1 = 2.54$ $N_1 = 0.1$  | <ul style="list-style-type: none"><li>✓ Ringing — decaying oscillation after the step</li><li>✓ Limit <math>\tilde{N}_1 = 0.75 = 3</math> quarter turns</li><li>✓ Over limit: cut <math>K_p</math> ↓, raise bands below ↑</li></ul> |
| $\Gamma_2$<br>$K_d$ band<br>$\Delta e$ vs $\Delta^2 e$ | Both portraits come from these two runs.                                          | <br>$N_2 = 2.58$ $N_2 = 0.29$ | <ul style="list-style-type: none"><li>✓ Buzzing — fast chatter on the transient</li><li>✓ Limit <math>\tilde{N}_2 = 1.00 = 4</math> quarter turns</li><li>✓ Over limit: cut <math>K_d</math> ↓, raise bands below ↑</li></ul>       |

## TUNING GUIDE (SPIN ITERATION)

|                                                                                            |                                                                                          |                                                                                           |                                                                                           |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>1</b><br><b>Step it</b><br>Close the loop with any gains, step the setpoint, record $e$ | <b>2</b><br><b>Trim it</b><br>Screen for runaway, cut where $ e $ last leaves 2% of peak | <b>3</b><br><b>Count it</b><br>Draw the three portraits, count turns<br>$N_0$ $N_1$ $N_2$ | <b>4</b><br><b>Move it</b><br>Use the table below to move $K_i$ $K_p$ $K_d$ , then repeat |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

| RULE — FIRST MATCHING ROW WINS | $K_i$   | $K_p$   | $K_d$   | READING                          |
|--------------------------------|---------|---------|---------|----------------------------------|
| record grows                   | ↘<br>÷2 | ↘<br>÷4 | ↘<br>÷8 | runaway — back off, rate hardest |
| $\Gamma_0$ loops               | ↓       | •       | •       | slow cycling — less reset        |
| $\Gamma_1$ loops               | ↑       | ↓       | •       | ringing — less gain              |
| $\Gamma_2$ loops               | ↑       | ↑       | ↓       | buzzing — less rate              |
| none loops                     | ↑       | ↑       | ↑       | all quiet — tighten              |

### TIPS

- ✓ Cut only the **lowest** objecting band — higher ones may be leakage
- ✓ Bands below the fault were just measured safe, so raise them
- ✓ Counts carry no units: the same limits fit every loop
- ✓ Any start works: the screen backs the gains off until the record settles
- ✓ No stopping test: stop at any iteration

↑ × y ↓ ÷ y ↓ back off ÷2 ÷4 ÷8 · hold |  $y = 1/(1-\beta)$ ,  $\beta = 0.1$ ;  $\tilde{N} = (0.5, 0.75, 1.0)$ ,  $\epsilon = 0.1$ ,  $\delta = 0.02$ ,  $\nu = 10$

Pachner, Otta, Dostál, Havlena — "Model-Free PID Tuning by Step-Response Inspection," arXiv:2609.15711, 2026.

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